

Zero Liquid Discharge (ZLD) Systems

Ultimate Water Recovery and Industrial Effluent Elimination.

Allianz Utilities Zero Liquid Discharge (ZLD) systems employ advanced thermal evaporation and crystallization technologies to eliminate wastewater discharge completely. The system purifies industrial effluent, recovering up to 98% of high-purity water for reuse while converting remaining dissolved solids into solid dry crystals for disposal or secondary resource recovery.

Key Features & Applications

KEY HIGHLIGHTS: - Complete elimination of wastewater discharge - High-efficiency mechanical vapor recompression (MVR) evaporators - Recovers up to 98% of process water for reuse - Robust corrosion-resistant material construction (Titanium/Duplex)	COMMON APPLICATIONS: - Chemical manufacturing - Pharmaceutical production - Power plants and mining - Eco-industrial zones
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System Process & Flow

Stage / Phase	Description
Pre-Concentration	Reverse Osmosis membranes concentrate salinity levels, reducing the volume sent to evaporators.
Thermal Evaporation	High-efficiency MVR heating evaporates water vapor, which is condensed and recovered as pure distillate.
Crystallization	The remaining highly concentrated brine slurry is crystallized, forming a dry solid cake of salt.
Distillate Recovery	Pure condensed water is returned directly to the plant's process loop, closing the water cycle.